ANDROID ENGINEERING / DESIGN-LED SYSTEMS

DEREK F. DILEO

Android Software Engineer

Kotlin · Jetpack Compose · UI Architecture

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LINKEDIN GITHUB DEREKDOTDEV.GITHUB.IO

01 PROFILE

Android software engineer and UI architecture owner guiding the incremental modernization of a production, hardware-integrated platform. Establishes team-wide architecture and testing practices using Kotlin, Jetpack Compose, dependency-inverted domain interfaces, and JNI-backed C++ integration. Shapes technical direction through design reviews, code review, and deliberate team enablement.

02 PROFESSIONAL EXPERIENCE

TOMAHAWK ROBOTICS

Acquired by AeroVironment (2023)

Software Engineer II, Applications

Jul 2025 – Present

Previously: Software Engineer I (Oct 2023 – Jul 2025) · Junior Software Engineer (Aug 2022 – Oct 2023)

KOTLIN / JAVA / JETPACK COMPOSE / MATERIAL 3 / MVI / HILT / ROOM / JNI / C++

- **Lead incremental Android modernization** from imperative Java/XML and LiveData toward Kotlin and Jetpack Compose, introducing clear presentation, domain, and data boundaries across new features and legacy UI migrations.
- **Shape architectural decisions across the team** through every UI and software design review and review of nearly all merged code; champion SOLID principles, dependency inversion, and reusable conventions that empower independent technical decisions.
- **Spearheaded a team-wide "new feature, new tests" initiative:** the team added **3,400+ Kotlin/JVM tests** in the past year; introduced the codebase's first Jetpack Compose UI tests, making refactoring safer and more routine.
- **Progressively isolate native C++ integration** behind data-layer implementations of domain interfaces, moving business logic into the domain layer while keeping Compose screens and ViewModels presentation-focused.
- **Architected operator-facing vehicle-pairing applications** that replace manual CLI configuration with automated discovery and configuration workflows for ground radios and vehicle nodes.
- **Built and deployed a React/TypeScript self-service portal on AWS GovCloud** for hardware registration, licensing, and 24/7 activation, reducing manual support work through pre-provisioned entitlements.
- **Automated customer issue reporting** with Jira API integration, eliminating manual bug-ticket entry.

02 PROFESSIONAL EXPERIENCE · CONTINUED

RX DELIVERED NOW

Quality Assurance Intern — Mar 2022 – Aug 2022

FLUTTER / DART / FIREBASE / ANALYTICS / JIRA

- Led QA across mobile and web applications, partnering with engineering and product to drive defects through resolution and implement analytics and conversion-tracking tools.

VALENCIA COLLEGE

School of Arts & Entertainment · Orlando, FL

Lab Assistant II · Systems & Dante Network Administration — Sep 2015 – Aug 2021

LINUX / DOCKER / CISCO NETWORKING / DANTE DOMAIN MANAGER

- Designed and deployed what became North America's largest educational RedNet network, using Linux virtualization and Dockerized infrastructure to support isolated virtual audio environments.
- Expanded a lab-support position into infrastructure ownership spanning systems administration, networking, technical documentation, production support, and supervision of student assistants.

03 SELECTED CAPSTONE

CITY OF ORLANDO · UNHOUSED YOUTH SOLUTIONS

Apr – Aug 2022

Technical Lead & Full-Stack Developer

JAVA / SPRING BOOT / SPRING SECURITY / MYSQL / AWS RDS

- Led a development team delivering a full-stack civic application, coordinating technical work, code reviews, and stakeholder requirements; designed its relational data model and persistence layer.

04 TECHNICAL SKILLS

ANDROID	Kotlin, Java, Jetpack Compose, Material 3, Android Views/XML, Coroutines/Flow, Room, Hilt, JNI/C++
ARCHITECTURE	Clean Architecture, MVI, MVVM, SOLID, dependency inversion, interface-driven design, UI/domain/data layering
TESTING / DELIVERY	JUnit, AssertK, Kotlin/JVM unit testing, Compose UI testing, instrumented testing, Gradle, Git, Jenkins
BACKEND / WEB	AWS GovCloud, SQL, REST APIs, Spring Boot, Hibernate/JPA, React, TypeScript, Flutter/Dart
INFRASTRUCTURE	Docker, Linux, networking, Jira, GitHub/GitLab

05 EDUCATION

VALENCIA COLLEGE

Orlando, FL

B.A.S. · Computer Engineering & Software Development — 2022 · With Honors

A.S. · Audio Engineering — 2020

A.S. · Sound Production — 2020